

A Model of Currency Linkages During Crises

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Outline

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Exchange Rates and Log-Transformation

- $S_{i,t}$ = units of currency i per USD ($i = 1, \dots, N$).
- An increase in $S_{i,t}$ indicates USD appreciation against currency i .
- Define

$$X_{i,t} = \log S_{i,t}, \quad X_t = (X_{1,t}, \dots, X_{N,t})'.$$

Notes:

- This transformation is standard, as log-prices simplify many econometric properties such as unit-root testing, cointegration analysis, and VECM estimation.
- Log-differences,

$$\Delta X_{i,t} = X_{i,t} - X_{i,t-1},$$

correspond approximately to percentage changes (returns).

From Present Value to One-Step SDF

- Continuous discounting:

$$p(0, t) = e^{-rt}, \quad p(0, t; X) = Xe^{-rt}.$$

- Discrete one-step pricing:

$$P_t = \mathbb{E}_t[M_{t+1}D_{t+1}].$$

- D_{t+1} is the payoff in USD at $t + 1$.
- M_{t+1} is the stochastic discount factor that brings $t+1$ payoffs back to t .
- Normalization: $\Delta t = 1$ and $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$.

Log-Normal SDF (USD)

$$\log M_{t+1} = -r_t - \underbrace{\frac{1}{2}\zeta_r^2}_{\text{Jensen correction}} - \xi_{t+1}$$

- r_t : USD short rate (one-period risk-free rate).
- ζ_r^2 : conditional variance of SDF innovations.
- Expectation condition: $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$, enforced by the Jensen.
- ξ_{t+1} : SDF innovation, modeled as

$$\xi_{t+1} = \lambda_r' g_{t+1} + \nu_{t+1}, \quad \mathbb{E}_t[g_{t+1}\nu_{t+1}] = 0.$$

- $g_t = (g_{1,t}, \dots, g_{q_r,t})'$: q_r independent $I(1)$ global factors
- $\lambda_r \in \mathbb{R}^{q_r}$: SDF factor loadings in regime r .
- $\nu_{t+1} \sim \mathcal{N}(0, \sigma_\nu^2)$: idiosyncratic USD pricing-kernel shock, orthogonal to g_{t+1} .

Factor Dynamics and SDF Innovation Variance

- Each factor follows a random walk with regime-specific drift and volatility:

$$g_{k,t+1} = g_{k,t} + \mu_{k,r} + \sigma_{g_{k,r}} \varepsilon_{k,t+1}, \quad \varepsilon_{k,t+1} \sim \mathcal{N}(0, 1).$$

- $\sigma_{g_{k,r}}$ is the volatility for factor k in regime r .
- It scales the standard normal shock $\varepsilon_{k,t+1}$, so volatility is tied directly to each factor and regime.
- These volatilities are collected in the diagonal covariance matrix:

$$\Sigma_{g,r} = \text{diag}(\sigma_{g_1,r}^2, \dots, \sigma_{g_{q_r},r}^2).$$

- The total SDF innovation variance is:

$$\zeta_r^2 = \lambda_r' \Sigma_{g,r} \lambda_r + \sigma_\nu^2.$$

- $\lambda_r' \Sigma_{g,r} \lambda_r$ represents the exposure from global risk factors.
- σ_ν^2 represents idiosyncratic risk.

Payoffs, No-Arbitrage, and Affine Solution

- A one-period claim pays $D_{i,t+1}$ units of currency i .
- Its USD price is:

$$P_{i,t} = \mathbb{E}_t \left[M_{t+1} \frac{D_{i,t+1}}{S_{i,t+1}} \right].$$

- $S_{i,t+1}$ is the exchange rate (units of currency i per USD).
- Dividing by $S_{i,t+1}$ converts the payoff into USD.
- Checking units explicitly:

$$\underbrace{D_{i,t+1}}_{\text{currency } i} \times \underbrace{\frac{1}{S_{i,t+1}}}_{\text{USD} / \text{currency } i} = \text{USD}.$$

- Thus, $D_{i,t+1}/S_{i,t+1}$ is the payoff expressed in USD terms.

FX Pricing Restriction and Economic Intuition

- No-arbitrage condition:

$$\frac{1}{S_{i,t}} = \mathbb{E}_t \left[M_{t+1} \frac{1}{S_{i,t+1}} \right].$$

- Equivalent form:

$$1 = \mathbb{E}_t \left[M_{t+1} \frac{S_{i,t}}{S_{i,t+1}} \right] = \mathbb{E}_t \left[M_{t+1} e^{X_{i,t} - X_{i,t+1}} \right].$$

- LHS: current USD cost of acquiring one unit of currency i , $\frac{1}{S_{i,t}}$.
- RHS: expected USD value of receiving one unit of currency i at $t+1$, discounted by M_{t+1} .
- Economic intuition: under no-arbitrage, today's USD cost of foreign currency equals its discounted expected future value. Any gap would imply a riskless arbitrage.

FX Pricing and Log-Normal Affine Solution

- The USD price of one unit of currency i delivered at $t + 1$ is:

$$\frac{1}{S_{i,t}}.$$

- No-arbitrage condition:

$$1 = \mathbb{E}_t \left[M_{t+1} e^{X_{i,t} - X_{i,t+1}} \right].$$

- Solving under log-normality and stationary fundamentals z_t yields the affine form:

$$X_{i,t} = \mu_{i,r} + \lambda'_{i,r} g_t + \Gamma'_i z_t + u_{i,t}. \quad (7)$$

- $\lambda_{i,r} \in \mathbb{R}^{q_r}$: currency-specific exposure to the q_r global factors $g_t \in \mathbb{R}^{q_r}$.

Autonomy in Factor Loadings

- Factor loadings for currency i in regime r are parameterized as:

$$\lambda_{i,r} = (1 - \psi_{i,r}) \odot \bar{\lambda}_i, \quad \bar{\lambda}_i \neq 0 \text{ fixed.}$$

- $\psi_{i,r} = (\psi_{i1,r}, \dots, \psi_{iq_r,r})' \in [0, 1]^{q_r}$: vector of autonomy parameters.
- $\bar{\lambda}_i \in \mathbb{R}^{q_r}$: baseline (regime-invariant) factor loadings.
- $\odot$: Hadamard (elementwise) product.
- Low $\psi_{ik,r}$: stronger pass-through from factor k (larger $|\lambda_{ik,r}|$).
- High $\psi_{ik,r}$: greater autonomy from factor k in regime r .

Assumption A1: Crisis Volatility

A1: For at least one factor k ,

$$\sigma_{g_k, \text{crisis}}^2 > \sigma_{g_k, \text{pre}}^2 \quad \text{and} \quad \sigma_{g_k, \text{crisis}}^2 \geq \sigma_{g_k, \text{post}}^2.$$

Crisis periods are characterized by heightened volatility in one or more global factors, reflecting:

- Increased uncertainty,
- Tighter global financial conditions,
- Or both.

Implications of A1

- Higher factor volatility $\sigma_{gk,r}$ makes factor shocks $\sigma_{gk,r}\varepsilon_{k,t+1}$ larger.
- This leads to more variability in the global factors g_{t+1} .
- Since $\xi_{t+1} = \lambda_r' g_{t+1} + \nu_{t+1}$, the SDF innovation itself becomes more volatile.
- The variance term

$$\zeta_r^2 = \lambda_r' \Sigma_{g,r} \lambda_r + \sigma_\nu^2$$

also rises, because $\Sigma_{g,r}$ has larger entries in crises.

- **Takeaway:** Crises amplify both the realized shocks (ξ_{t+1}) and the expected variance (ζ_r^2) of the SDF. In crises, higher SDF volatility means stronger discounting of risky cash flows, leading to cheaper asset prices and higher risk premia.

Assumption A2: Autonomy Compression

A2: For all i, k ,

$$\psi_{ik,\text{crisis}} \leq \psi_{ik,\text{pre}}.$$

Autonomy compresses during crises as market stress and funding pressures force more synchronized exchange-rate movements with the global factors.

Implications of A2

- Lower autonomy $\psi_{ik,r}$ means stronger pass-through from global factors.
- This raises the effective loadings $\lambda_{i,r}$ on the global factor vector g_{t+1} .
- With larger $\lambda_{i,r}$, the conditional variance

$$\zeta_r^2 = \lambda_r' \Sigma_{g,r} \lambda_r + \sigma_\nu^2$$

increases.

- The SDF innovation $\xi_{t+1} = \lambda_r' g_{t+1} + \nu_{t+1}$ also becomes larger in magnitude.
- **Takeaway:** In crises, reduced autonomy forces currencies to move more tightly with global shocks, amplifying both SDF variance and innovations.

Factor-Loading Structure Across Regimes

- For each regime r , define the $N \times q_r$ factor-loading (pass-through) matrix:

$$\Lambda_r = \begin{bmatrix} \lambda'_{1,r} \\ \vdots \\ \lambda'_{N,r} \end{bmatrix} = \begin{bmatrix} (1 - \psi_{11,r})\bar{\lambda}_{11} & \cdots & (1 - \psi_{1q_r,r})\bar{\lambda}_{1q_r} \\ \vdots & \ddots & \vdots \\ (1 - \psi_{N1,r})\bar{\lambda}_{N1} & \cdots & (1 - \psi_{Nq_r,r})\bar{\lambda}_{Nq_r} \end{bmatrix}.$$

- Each row $\lambda'_{i,r}$ corresponds to currency i 's loadings in regime r .
- These rows are parameterized elementwise as:

$$\lambda_{i,r} = (1 - \psi_{i,r}) \odot \bar{\lambda}_i, \quad \bar{\lambda}_i \neq 0 \text{ fixed.}$$

- $\psi_{i,r}$ are autonomy parameters controlling how strongly currency i is tied to global factors in regime r .

Cointegration Rank and Regime Dependence

Given the affine system:

$$X_t = \mu_r + \Lambda_r g_t + \Gamma z_t + u_t,$$

the long-run comovement among N exchange rates in regime r depends on the rank of the pass-through matrix Λ_r .

Key result:

$$\text{rank}(\Lambda_r) \leq \min(N, q_r).$$

Interpretation:

- $\text{rank}(\Lambda_r) = 1$: single nonstationary common component; $N - 1$ cointegrating relations (single-factor baseline).
- $\text{rank}(\Lambda_r) = k > 1$: $N - k$ cointegrating relations (multiple global factors).
- $\text{rank}(\Lambda_r) = N$: no cointegration (all trends common, cannot form stationary relations).

Special Case: $\text{rank}(\Lambda_r) = 0$

- Algebraically, cointegration rank is

$$N - \text{rank}(\Lambda_r) = N.$$

- In this case there are no common stochastic trends.
- If all trends are idiosyncratic, each currency follows its own path with no shared driver, so no stationary linear combinations exist.
- If all variables are stationary ($I(0)$), then they already revert to a mean and cannot be cointegrated.

Stationarity, $I(1)$ and Cointegration

- $I(1)$ variables are non-stationary random walks. They drift without mean reversion.
- Cointegration means there exist linear combinations of $I(1)$ variables that eliminate stochastic trends.
- These combinations are $I(0)$ — stationary and mean-reverting.
- Interpretation: exchange rates may wander individually, but relative to each other they share long-run equilibrium relations.

Regime Variation

- Factor volatilities $\sigma_{gk,r}^2$ and autonomy parameters $\psi_{ik,r}$ vary across regimes, changing the effective rank of Λ_r .
- Crises can increase similarity in response to global factors, making common trends more dominant.
- Crises can also compress autonomy parameters, aligning currencies more closely with global factors.
- New factors (e.g., pandemic-specific shocks) can raise autonomy, reduce cointegration, or even eliminate it if enough independent trends appear.

Proposition 1: Rank–Cointegration Link

$$\text{Cointegration rank} = N - \text{rank}(\Lambda_r)$$

- Cointegration rank is the number of stationary long-run linkages among the N exchange rates.
- If $\text{rank}(\Lambda_r) = 1$: all persistent variation comes from a single global factor (e.g., USD funding/liquidity). This implies that the system has $N - 1$ cointegrating relationships.
- As $\text{rank}(\Lambda_r)$ increases: more independent global trends drive the system, reducing the number of cointegrating relations.
- If $\text{rank}(\Lambda_r) = N$: each currency follows its own independent stochastic trend, no common trends remain, which implies no cointegration (no stable long-run relationships).

Proposition 2 (Crisis Co-Movement Amplification)

If $\text{rank}(\Lambda_r)$ remains unchanged in crisis, but (i) $\sigma_{gk,r}^2$ rises for one or more factors, and (ii) $|\lambda_{ik,r}|$ converge across currencies (autonomy compression), then long-run correlations increase.

Economic interpretation: Crises amplify co-movement through two reinforcing channels:

1. Variance channel

Higher volatility in the common factors increases their share of total long-horizon variance.

2. Loading-alignment channel

Lower dispersion in $\psi_{ik,r}$ makes currency responses more similar in magnitude and sign, reducing idiosyncratic offsets.

Summary: Crisis Co-Movement Drivers

Two reinforcing drivers of crisis co-movement:

- **Increased factor volatility:** Larger $\Sigma_{g,r}$ boosts the relative weight of common factors versus idiosyncratic shocks, magnifying the importance of global drivers.
- **Stronger alignment of loadings:** As $\lambda_{i,r}$ and $\lambda_{j,r}$ become more similar, pairwise correlations tighten toward one, increasing co-movement across currencies.

Proposition 3: Rank Change in Crisis

If new persistent global shocks emerge, or if autonomy shifts occur (e.g. a new idiosyncratic stochastic trend), then:

$$\text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{non-crisis}}).$$

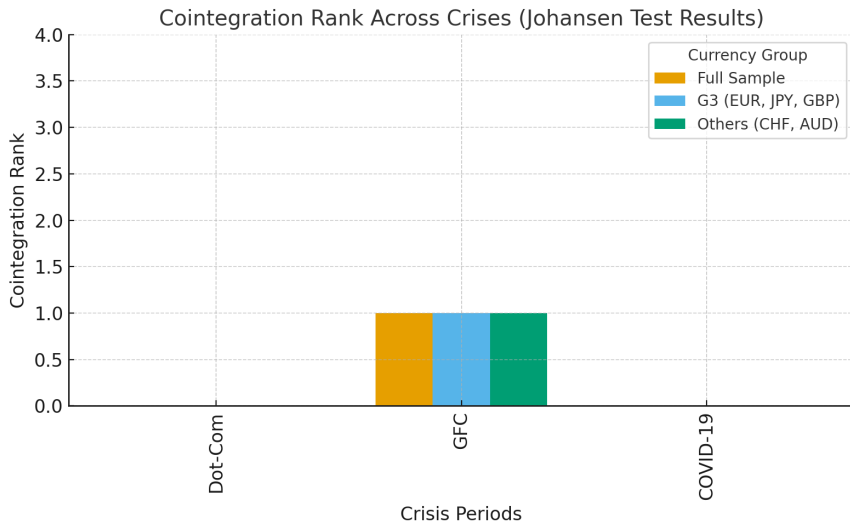
- This increases the number of independent stochastic trends driving the system.
- As a result, the number of cointegrating relationships falls.
- In the extreme case:

$$\text{rank}(\Lambda_{\text{crisis}}) = N,$$

cointegration disappears entirely (no long-run equilibrium remains).

- **Long-run links (rank):** Does a stable cointegrating anchor exist in each crisis?
- **Who adjusts (VECM α):** Which currencies bear the error-correction burden in the GFC?
- **Rolling long-horizon correlations (H=180):** Do correlations approach 1?
- **Crisis intensity (JPMVXYGL) → correlations:** Do high-vol regimes compress autonomy?

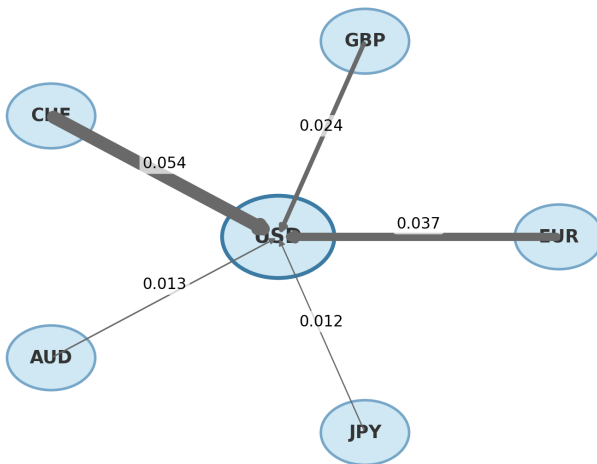
Cointegration rank across crises



Dot-Com and COVID-19: rank = 0; GFC: rank = 1 (< 4).

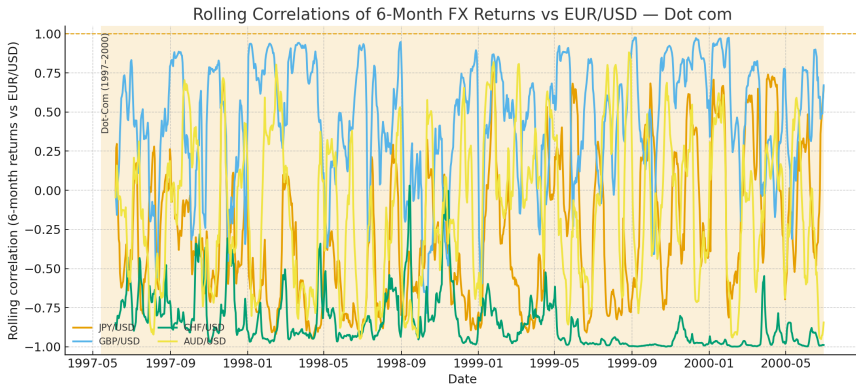
Who adjusts? VECM α during GFC ($r=1$)

VECM Adjustment Speeds (GFC, rank=1)



EUR/CHF/GBP adjust (significant α); AUD/JPY weakly exogenous.

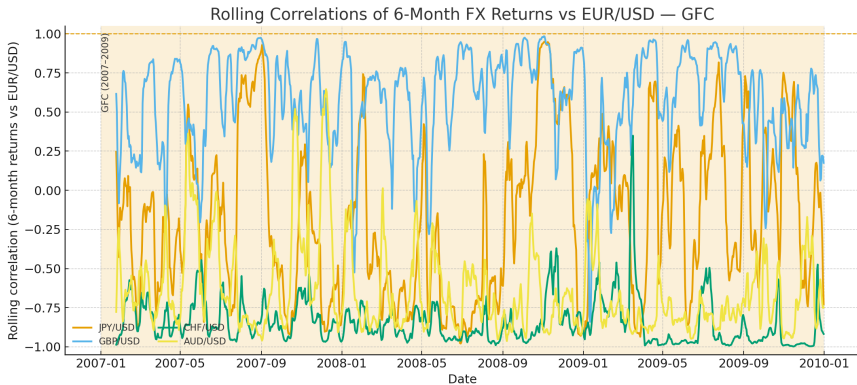
Rolling 6-month correlations vs EUR/USD — Dot-Com



Most of lines remain dispersed/near 0 or negative; no persistent move toward 1.

No Convergence

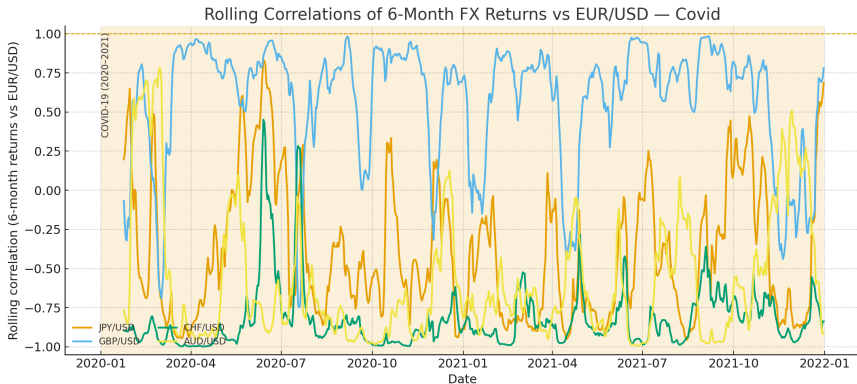
Rolling 6-month correlations vs EUR/USD — GFC



Notice EUR–GBP near 0.75–0.9, not universal alignment.

Partial Convergence

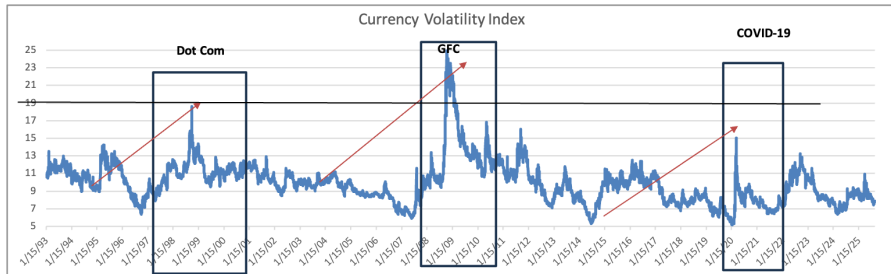
Rolling 6-month correlations vs EUR/USD — COVID-19



Volatile, often near/ below 0; no approach to 1.

No Convergence

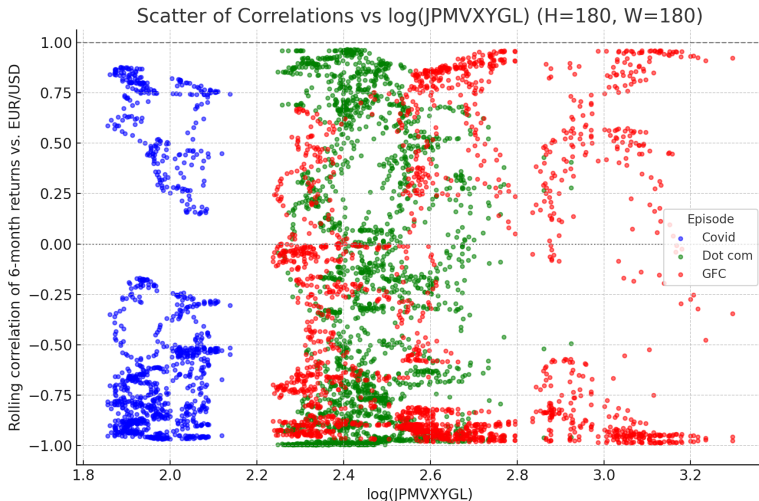
JP Morgan Global Currency Volatility Index (level)



Crisis windows highlighted; volatility spikes in all three episodes. ¹

¹(used in scatter analysis next slide)

Correlations vs $\log(\text{JPMVXYGL})$ ($H=180$)



GFC shows high volatility $\Rightarrow$ partial convergence; **Dot-Com** & **COVID-19**: high volatility without convergence.

Thank You for Listening!

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Questions?